

Computing 1

First steps with MCMC sampling

All code listed here, is available at URL
<https://coursesupport.physics.otago.ac.nz/wiki/pmwiki.php/ELEC446/BUC5>.

Random walk Metropolis sampling from $N(0, 1)$

Here is Matlab code for a random-walk Metropolis targeting $N(\mu, \sigma^2)$, having pdf (probability density function)

$$N(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}.$$

```
function X=mcgaus(mu,sig,x0,N,w)
%function X=mcgaus(mu,sig,x0,N,w)
%
% Return N samples of N(mu,sig^2) using a
% RWMH with proposal window w, starting at x0

X = zeros(1,N);
X(1)=x0;

for k=1:(N-1)
    %xp = X(k)+w*(2*rand-1); % uniform window
    xp = X(k)+w*randn; % normal window, variance w^2

    alpha = min(1,exp( -(xp-mu)^2+(X(k)-mu)^2)/(2*sig^2) ));

    if rand<alpha
        X(k+1)=xp;    % accept
    else
        X(k+1)=X(k); % reject
    end
end
```

Notes:

- This is a Metropolis algorithm (as opposed to Metropolis–Hastings) because it uses a *symmetric* proposal, so the Hastings ratio is not needed (as it equals 1).
- The Metropolis ratio is calculated using the difference of *logarithms* of pdfs, then exponentiates, i.e., as

$$\frac{N(x^*; \mu, \sigma^2)}{N(x; \mu, \sigma^2)} = \exp \{ \log N(x^*; \mu, \sigma^2) - \log N(x; \mu, \sigma^2) \}$$

to avoid numerical issues. You will investigate the numerical problems created by evaluating the Metropolis ratio directly as a ratio of pdfs.

Exercises:

1. Run the code in `mcgauss.m` to generate a chain of length 10^4 that targets $N(0, 1)$. Use windows $w = 0.3, 3, 30$.
2. Plot histograms of each chain. Plot the chains, and see which window gives the most efficient mixing. (Code is in `mcgauss_demo.m`.)
3. Adapt `mcgauss.m` to sample from the exponential distribution with mean $1/\lambda$, having pdf

$$\text{Exp}(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Calculate the Metropolis ratio in the logarithm, and set $\lambda = 1$. You can deal with the pdf being zero (for $x < 0$) in two ways: allow $\log 0 = -\infty$ and evaluate the acceptance probability as in the code, to reject proposals $x^* < 0$, or simply reject in the proposal, and – important – advance the chain one step repeating state x . (rejecting in the proposal is more efficient when evaluating the acceptance probability is expensive). See code in `mcmc.m`.

- Start at $x = 1$, calculating acceptance probability in the logarithm (numerically stable) to check that everything works OK.
 - Now start at $x = 1000$, plot histograms, and see that everything is still OK.
 - Now start at $x = 1000$ and use the numerically unstable version of evaluating the ratio of probabilities to see the effect of numerical precision.
4. Adapt `mcgauss.m` to sample from the *discrete* geometric distribution for $p \in [0, 1]$, having pmf

$$\text{Geom}(k; p) = (1 - p)^k p \quad k = 0, 1, 2, 3, \dots$$

Of course you will need to modify the proposal in `mcgauss.m`. Run chains and verify that the mean of $\text{Geom}(k; p)$ is $(1 - p)/p$.

Congratulations! Now you are an expert.